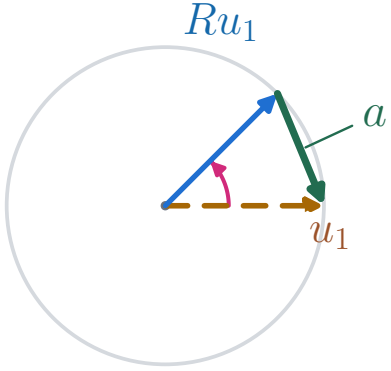
$U = \mathrm{span}\{e_0, e_1\}, \quad V = R(U) = W(U), \quad \theta_1 = \theta_2 = 45°$

Direct rotation R

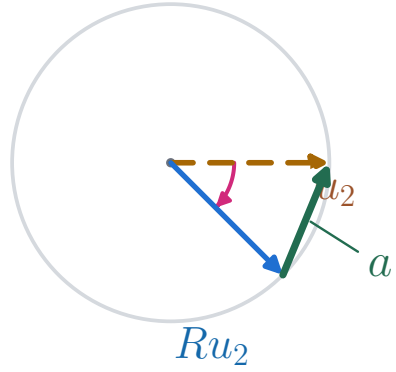
$\Pi_1 \oplus \Pi_2 = \mathbb{R}^4$ (principal planes)

$\Pi_1 = \mathrm{span}\{e_0, e_3\}$



$+45°$

$\Pi_2 = \mathrm{span}\{e_1, e_2\}$

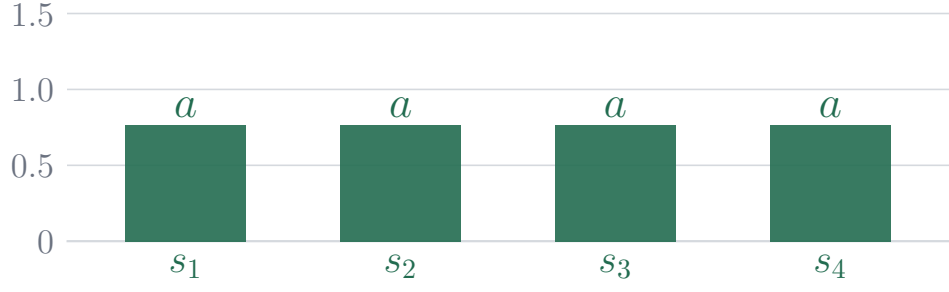


$-45°$

Unit input q — Image Tq

Singular values of $I - R$

$a = \sqrt{2 - \sqrt{2}} \approx 0.765$



from Π_1 from Π_2

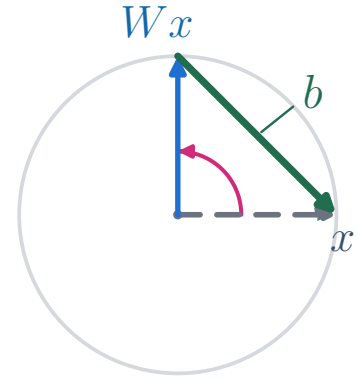
$\|I - R\|_1 = 4a \approx 3.061$

$>$

Competitor W

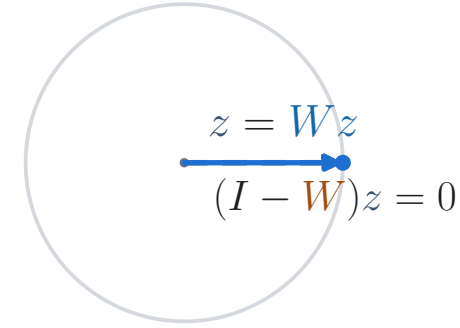
$M \oplus Z = \mathbb{R}^4$ (moving and fixed planes)

$M = \mathrm{range}(W - I)$



$+90°$

$Z = \ker(W - I)$

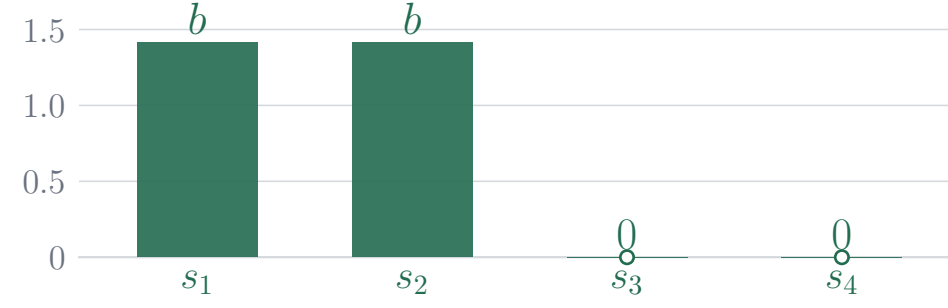


$0°$ (fixed)

Displacement $(I - T)q$

Singular values of $I - W$

$b = \sqrt{2} \approx 1.414$



from M from Z

$\|I - W\|_1 = 2b \approx 2.828$